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Towards a methodologically congruent framework for GenAI use in nonpositivist qualitative research

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ABSTRACT

GenAI is increasingly used in qualitative research, yet much current practice remains methodologically incongruent. Attempts to make LLMs replicate qualitative coding often create a mismatch between underlined qualitative values (e.g. depth, context, and reflexivity) and actual AI-supported analytic practices (e.g. surface-level extraction and automated coding). This article distinguishes between technological incongruence, where GenAI is asked to perform tasks it is not designed to handle, and qualitative methodological incongruence, where AI workflows reproduce the surface procedures of qualitative analysis while neglecting its interpretive and reflexive commitments. We propose a shift from 'AI as coder' to 'AI as interlocutor'. We outline key tenets for methodologically congruent GenAI use in non-positivist qualitative research: relational epistemology, ethical, responsible and rigorous use, researcher's lead role, and abductive dialoguing. GenAI can support qualitative analysis by surfacing, retrieving, comparing, and suggesting material for interpretation, but meaning-making remains the responsibility of the human researcher.

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Large language models; methodological (in) congruence; nonpositivist qualitative research; qualitative data analysis; thematic analysis; Generative AI

1. Introduction: the methodological crossroads

The field of qualitative data analysis (QDA) has encountered a turning point. Since the emergence of ChatGPT in November 2022, there have been hundreds of Generative AI (GenAI) tools released. Apart from ChatGPT, at the time of this writing, the latest AI models are Gemini 3.5 by Google, GPT 5.6 by OpenAI, Grok 4.3 by xAI, DeepSeek 4 by DeepSeek, Claude Opus 4.8 and Fable 5 by Anthropic, Llama 4 by Meta, to name just a few.

The rise of GenAI and LLMs, designed to process and generate text, has created a growing body of scholarship that examines the applications of these technologies in QDA. In terms of technological systems through which GenAI/LLMs are used in QDA, three main user-facing approaches have emerged(Nguyen-Trung & Nguyen, 2026; Silver, 2023): (1) Incorporating GenAI functions into existing computer-assisted qualitative data analysis software (CAQDAS) packages such as ATLAS.ti, MAXQDA (AI Assist),

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NVivo (AI Assistant) or QDA Miner. (2) Developing or using AI applications purpose-built for qualitative research, including commercial platforms such as Tailwind, QInsights, or AILYZE, as well as bespoke or open-source applications (e.g., TALLMesh). (3) Using AI chatbots (general-purpose AI) such as ChatGPT, Claude, NotebookLM or Gemini. Across these approaches, the underlying language models may be accessed or deployed in different ways: through APIs connected to externally hosted proprietary models; by locally self-hosting open-weight models, such as Llama and Mistral, using tools such as Ollama, LM Studio, or GPT4All; or through models deployed on institution-controlled infrastructure.

The dominant methodological trend has been an attempt to replicate human coding behaviour (Dunivin 2025; Etesse, 2024; Firetto et al., 2025; Fischer, 2025; Flanders et al., 2025; Gao et al., 2023; Gustavsen et al., 2025; Lee et al., 2023; Şen et al., 2023; Turobov et al., 2024; Zackary, 2024, among many others). Our review of this literature suggests that this 'replication approach', driven by automation ambition, often leads to significant methodological incongruence. This concern may partly explain the emergence of a rejection movement led by the open letter signed by 417 scholars from 31 countries published in Qualitative Inquiry (Jowsey, Braun et al., 2025) that rejects GenAI use in reflexive qualitative research. Many follow-up works in this movement criticised the uncritical adoption of GenAI in qualitative research, but they often stopped at epistemological/theoretical rejection and never discussed what critical, methodological congruent integration of GenAI looks like (such as Griep et al., 2026; Levitt, 2026; Nguyen & Welch, 2026; see the review by Nguyen-Trung et al., 2026). By contrast, a parallel position has emerged in support of the responsible and critical integration of GenAI in qualitative research, led by Friese et al. (2026), endorsed by 114 scholars from 22 countries. Alongside this position, there is a growing body of work reporting use cases for GenAI in qualitative research (Brailas & Katerelos, 2025; Wise et al., 2026; Nguyen-Trung, 2025; Nguyen-Trung & Nguyen, 2026). However, the question of how LLMs fit meaningfully into qualitative research is rarely addressed.

To close the gap, this article's aim is twofold. First, based on Clarke et al. (2024), we focus on reframing two distinct but equally problematic practices of the methodological misalignment: technological incongruence and qualitative methodological incongruence. Second, inspired by Tracy (2010), we propose a set of key criteria for nonpositivist researchers to ensure their GenAI application is methodological congruence.

2. Extending discussions on methodological incongruence

Clarke et al. (2024), drawing on Hills (2000), define methodological congruence as 'the alignment between all aspects of research from philosophical orientation through to data generation methods' (p. 10). Incongruence, then, occurs when there is a gap between the values a researcher claims to uphold and the actual practices they employ. For example, a researcher might state, 'I am doing constructivist qualitative research', while simultaneously describing work in ways that contradict this stance, such as claiming, 'AI helped me automate the entire data analysis process.' Thus, methodological congruence first comes from the self-awareness and theoretical sensibility that help researchers know 'how paradigms and methods align' (Tracy, 2026, p. 252).

Yet, it is not rare that researchers mix or combine elements from different methods or approaches. On 'meaningful coherence' – her concept for methodological congruence, Tracy (2010, p. 848) remarks:

> meaningful coherence does not mean that a study cannot borrow or use concepts from multiple paradigms (a key part of crystallization). Indeed, some studies are creative or groundbreaking precisely because they borrow from various theories or frameworks to create something new.

However, this mixing needs reasonable 'meaningful' rationales. In Clarke et al. (2024, p. 10) own words, incongruence also happens when '(perhaps unknowingly) combining incommensurate frameworks … without any explanation'. Tracy (2010, p. 848), again, recommends 'To catch and hold the attention of key audiences, it is important to include a clear purpose statement early on in the piece and diligently attend to that statement'. Then, the methodological incongruence will arise when researchers cannot explain convincingly *why* they chose to or *how* they did to mix different methodological elements. To provide this explanation, researchers must be aware of the methodological values shaping their decisions or choices of methods, and be clear of why it is necessary to bring different values into one single study. When researchers fail to resolve the tensions between these frameworks, their study lacks an appropriate philosophical foundation, undermining both the integrity of the analysis and the trustworthiness of its conclusions.

2.1. Technological incongruence

Firstly, we observe technological incongruence, where qualitative researchers explore the use of LLMs for established frameworks like thematic analysis (TA), grounded theory, or content analysis without sufficient understanding of GenAI's technological capabilities (Jowsey, Stapleton et al., 2025; Mayring, 2025; Mesec, 2023; Nguyen & Welch, 2026; Theelen et al., 2024; Wachinger et al., 2024). Not every task can be outsourced to AI/LLMs simply because they are not designed to do so.

Ignoring or overlooking GenAI's technical capabilities, these researchers often employ GenAI/LLMs using basic prompts to replicate manual coding steps without human interpretation. For instance, Sakaguchi et al. (2025) used ChatGPT to analyze data in three ways, including general thematic approach, grounded theory, and five-step framework. In their words, they wrote (Sakaguchi et al., 2025, p. 3): 'Please act like a researcher with expertise in qualitative research and analyze the transcript I have provided you with following the grounded theory approach proposed by Charmaz'. The problem is with just one single, one-off prompt, they expect LLMs to conduct a grounded theory analysis of the whole dataset. In another example, Friedman et al. (2025, p. 1066) tested ChatGPT and Gemini's assessment of content meaning by asking them to summarize 'what this quote means: "If two guys want to get it. If two girls want to get it"'. They found both chatbots' responses lacked nuance and concluded that these LLMs are vulnerable to 'hallucinations', but they never questioned whether, if they had provided more contextual information around the quote, the chatbots' performance would have been better.

In another example, to test GenAI's capability, Jowsey, Stapleton et al. (2025, p. 5) used one single, one-turn, zero-shot prompt to guide Microsoft Copilot to conduct the entire

Table 1. Analysis of Jowsey, Stapleton et al. (2025) prompt.

Phase/tasks	Original prompt text by Jowsey, Stapleton et al. (2025) (with task numbers added)	Critical questions (ambiguities & limitations)
Overall task (1.0)	Undertake thematic analysis of the dataset I have uploaded alongside this prompt.	• What type of TA (reflexive TA, template analysis, framework analysis or coding reliability TA; inductive or deductive) should Copilot use?
1. Familiarisation (tasks 1.1–1.3)	[1.1] Review the dataset in its provided format. Begin by [1.2] familiarising yourself with the content and [1.3] noting key recurring ideas.	• Review for what purpose or criteria? • Who is 'yourself'? Can Copilot have positionality? • Where does/How can it store 'recurring ideas' for subsequent steps? (no, it cannot)
2. Coding (tasks 2.1–2.2)	[2.1] Generate initial codes based on identified features and [2.2] apply these codes consistently across the dataset.	• How many transcripts should Co-pilot use to generate initial codes? • How can Co-pilot ensure consistency across the dataset without persistent memory? (no, it cannot)
3. Theme development (task 3.1)	[3.1] Group related codes into potential themes, using pattern detection algorithms if applicable.	• Can the Co-pilot store/memorize codes in order to group them into themes? (no, it cannot) • Are patterns detected by an algorithm equivalent to analytically derived themes? (no)
4. Theme refinement (tasks 4.1–4.4)	[4.1] Evaluate and [4.2] refine the themes to ensure they are coherent and relevant. [4.3] Develop detailed descriptions and [4.4] clear names for each theme.	• Can Co-pilot memorize themes generated earlier in order to evaluate and refine them? (no, it cannot) • What criteria define coherence and relevance? How can Co-pilot know these without researcher input?
5. Write-up (tasks 5.1–5.3)	[5.1] Prepare a comprehensive report summarising the thematic findings [5.2], supported by relevant data excerpts [5.3].	• What form should the report take? • What are the criteria for selecting data excerpts? • How do we ensure Co-pilot does not cherry-pick data excerpts?
6. Validation (tasks 6.1–6.3)	[6.1] Ensure consistency in coding and theme development, [6.2] validate the accuracy of themes, and [6.3] document the entire process transparently.	• Can Co-pilot memorize all themes, codes, and their links in order to ensure consistency? (no, it cannot) • What criteria define the accuracy of themes? • How can Co-pilot document the whole process without a persistent memory of all steps? (it cannot do this)
4. Theme refinement (tasks 4.5–4.8)	[4.5] Provide between four and eight themes, [4.6] supporting the themes with the number of participants that are represented in each theme. [4.7] Provide quotes from participants to support each theme. [4.8] Provide this thematic analysis summary within 800 words.	• What if the number of themes exceeds eight? On what criteria should the Co-pilot prioritize? • How can Co-pilot count exactly the number of participants represented in each theme? • Task 4.7 repeats 5.3 as both request supporting evidence (quotes/excerpts).

Source: Authors.

TA. In Table 1, we reanalyzed their prompt to identify phases, tasks, and hidden assumptions that hamper GenAI's performance. Column 2 provided original prompt text by Jowsey, Stapleton et al. (2025) and our identified task numbers without changing the prompt order.

The prompt assumes that the chatbot can conduct TA as a sequential analytic process in which the outputs of one stage become the inputs of the next. For instance, the result of Task 2.1 'Generate initial codes based on identified features' is a list of

codes that Co-pilot was expected to store and use to 'apply these codes consistently across the dataset' (Task 2.2). Similarly, Co-pilot was expected to store the revised list of codes after coding the whole dataset so as to conduct Task 3.1 'Group related codes into potential themes, using pattern detection algorithms if applicable'. This assumption does not match how consumer-facing chatbots operate. Chatbots generate responses to prompts but do not create a persistent analytic representation of the dataset. Consequently, codes, coded segments, themes, and supporting evidence are not maintained as stable analytic objects that can be systematically reviewed, retrieved, and revised throughout the analysis process. What Co-pilot likely did was not execute 21 analytic tasks sequentially. It probably recognized the overall pattern: 'User wants a thematic analysis report with 4–8 themes, participant counts, quotes, and a maximum of 800 words.' And then generated the kind of output that usually follows such a request.

The reported results are largely predictable: hallucinations, shallow descriptive codes (e.g. 'Health issues faced by adolescents,' 'Role of Peer Educators'), and outputs that shift unpredictably. Jowsey, Stapleton et al. (2025) report that Copilot drew most of its themes and quotes from the first two to three pages of the textual data, rather than from the full dataset. This suggests that Copilot was not working with a stable and complete representation of all documents when generating the final output. Instead, it likely relied on the most accessible parts of the uploaded material, together with a compressed or summary-level representation of the remaining content. This helps explain why quotes were fabricated. Once Copilot had generated themes, it was also asked to support them with participant quotes. However, if the relevant verbatim excerpts were not available in a retrievable form at the moment of generation, the model could still produce text that sounded like plausible participant evidence. In such cases, the quote is not retrieved from the dataset but reconstructed as an illustrative statement that fits the theme.

No single prompt, let alone one single zero-shot prompt with over 20 subsequent tasks, can get all the data analysis done. As we analyzed in Table 1, each task of TA requires researchers to intervene and address arising problems. This means, in each of the TA stages/activities, researchers need to prompt iteratively, going back and forth between data, their own input – analysis – reflection – verification, and LLM-generated outputs, to get desired results (Nguyen-Trung, 2025). Reanalyzing one of the five datasets reported by Jowsey, Stapleton et al. (2025), De Paoli (2026) found that the shallow results (e.g. lack of themes, ignoring overlapping between groups) reported by Jowsey, Stapleton et al. (2025) was because of fallacy of their designing. What De Paoli did differently was to divide the TA process into smaller steps to manage the LLM-generated outputs better: (i) setting up GPT-4o via Azure API, configuring the LLM parameters such as temperature (i.e. the model's predictability or creativity in its responses) and top-p (i.e. the possible next tokens the model is allowed to predicted); (ii) using a zero-shot prompting, but dividing this prompting into multiple iterations, each iteration working on creating initial codes for one interview only; (iii) after repeating the procedure and coding the whole dataset, asking the model to compare between lists of codes from different interviews to de-duplicate and create the final list of initial codes; (iv) asking the model to create themes based on the final list of codes. In short, the key to effective prompting here is dividing the process into small tasks, then work iteratively on each of these small tasks until all the TA process is done.

In sum, rather than recognizing that these outcomes stem from a prompt that was never technically feasible (Friese et al., 2026), the researchers perceived this not as a misalignment of *frameworks*, but as a failure of the *tool*. This resulted in a backlash, using these predictable failures as an argument to reject GenAI entirely, rather than interrogating the flawed workflow itself (Jowsey, Braun et al., 2025; Nguyen & Welch, 2025).

2.2. Qualitative methodological incongruence

2.2.1. Unknowing use of GenAI in Big Q qualitative research

In our review, many studies applying GenAI in QDA since 2022 have been authored by computer scientists, or quantitatively/technically trained researchers rather than qualitative experts (Anakok et al., 2025; Gao et al., 2023; Xiao et al., 2023; Zackary, 2024). These studies view qualitative data analysis (QDA) as 'inherently problematic' because this process is 'slow, labor intensive, exhausting, and expensive functioned' so that GenAI use is considered as 'a solution' (Paulus et al., 2025, p. 5). Then, their applications tend to develop a AI-powered/autonomous mechanical procedure to optimize QDA. They adopt LLMs as a shortcut to bypass the time-consuming work of coding, creating procedures to 'automate' analysis (Anakok et al., 2025; Dunivin, 2025; Gao et al., 2023; Katz et al., 2024; Zhang et al., 2023). While their intention is to help qualitative researchers – many of whom lack the technical skills to build such pipelines or connect to APIs as mentioned above – these technically oriented authors may never have engaged in manual coding themselves. While technically sophisticated, these technical-heavy approaches often reproduce surface-level procedures of existing qualitative methodologies *without* engaging with the interpretive values that give those procedures meaning. One of the clearest examples of this methodological incongruence occurs when studies claim to use Big Q approaches such as reflexive TA, as defined by Braun and Clarke (2019), while using GenAI or LLMs to carry out small q practices such as counting codes or themes and applying interrater agreement checks. We categorize this as a lack of qualitative literacy: an insufficient awareness of the different paradigms that underpin qualitative research practices and of the assumptions, values, and quality criteria associated with each tradition.

The unknowing use of GenAI might come from the unawareness of the differences in philosophical assumptions between different qualitative traditions that require well-versed training on histories, theories, and research methodologies. With a thorough understanding of this knowledge, researchers will become sensitive to the boundaries demarcating different traditions such as small q, Big Q, and medium Q (see Braun & Clarke, 2023; Kidder & Fine, 1987). 'Big Q' refers to the 'fully qualitative' traditions (e.g. reflexive TA (Braun & Clarke, 2019, 2022), phenomenology (Moran, 2002), constructivist grounded theory (Charmaz, 2006) that rests on non-positivist/relativist assumptions that emphasise partial and contextual, subjective, reflexive nature of knowledge production. 'Small q' refers to the qualitative tradition that uses the (post)positivist worldview to quantify data outputs (keywords, codes, themes), using interrater agreement check to avoid researchers' biases so as to ensure the finding of 'objective' truth (e.g. coding reliability TA (Boyatzis, 1998; Fereday & Muir-Cochrane, 2006; Guest et al., 2012). In between these two poles, there is a vast of medium Q, pragmatic, broadly non-positivist

methods that acknowledge reflexivity and subjectivity while employing a more structured approach, i.e. having a priori theoretical framework to guide the analysis. Examples are: Template Analysis (Brooks & King, 2014; King et al., 2018), matrix analysis (Miles & Huberman, 1994; Miles et al., 2014), two-cycle coding analysis (Saldaña, 2016); Framework Analysis (Gale et al., 2013; Ritchie & Spencer, 1994); Iterative Thematic Inquiry (ITI) (Morgan & Nica, 2020)). Knowing the boundaries among these traditions, positioning one's research within one appropriate tradition, or providing reasonable explanations when there is a need to traverse across different traditions is crucial to maintain methodological congruence.

Due to the mentioned unknowingness, many applications of GenAI engaged with Big Q approaches mechanically. For instance, some studies apply LLMs as 'an independent coder' to conduct grounded theory to produce 'in-depth and diversified codes' (Yang & Ma, 2025, p. 2526) without actually engaging with the central element of this methodology, which is the constant comparative method (Glaser & Strauss, 1967). This method considers coding in particular and data analysis in general not as an isolated activity, but in close association with theoretical sampling and data collection. This means coding does not just serve the purpose of labelling and organizing data, but more importantly, it needs to inform researchers about the need to diversify their data, which motivates researchers to conduct subsequent theoretical sampling and data collection stages. In Glaser and Strauss' (1967, p. 109) own words:

> By joint collection and analysis, the sociologist is tapping to the fullest extent the in vivo patterns of integration in the data itself; questions guide the collection of data to fill in gaps and to extend the theory-and this also is an integrative strategy.

In this constant comparative process for theory-building purposes, researchers are intensively required to link codes, categories, and their dimensions, then develop questions that lead to further data collection. Thus, it is an iterative process spread over time oscillating between data collection and analysis, which cannot be simulated by giving an LLM a dataset and a prompt and asking it to 'conduct' a grounded theory analysis. This can only lead to superficial results of grounded theory.

Another clear example of the unknowingness is the fact that many studies cite Braun and Clarke (2006) to justify their AI-driven theming (Anakok et al., 2025; Drápal et al., 2023; De Paoli, 2024) yet fail to engage with the method's evolution into a value-based, fully qualitative paradigm. In 2019, Brand and Clarke rebranded their TA as 'reflexive Thematic Analysis' (reflexive TA) (Braun & Clarke, 2019; Braun et al., 2019). This act does not merely give their approach a new name but indeed introduces stricter value-based guidelines that urge practitioners and researchers not to breach them to avoid methodological incongruence (Clarke et al., 2024). One of the examples of obvious incongruence when mechanically applying reflexive TA is quality insurance. Many studies attempt to validate AI coding or theming by checking inter-rater reliability between the human and the AI and calculating inter-rater reliability (IRR) (Dunivin, 2025; Nyaaba et al., 2025; Prescott et al., 2024; Wang et al., 2025; Wen et al., 2025). The practice creates a paradox: frameworks like reflexive TA or grounded theory explicitly reject IRR, arguing that coding is subjective and interpretive, not a search for a single 'correct' truth. By using IRR to 'prove' the AI works, researchers are imposing a (post) positivist metric onto a constructivist process – a textbook definition of methodological

incongruence. Although some might argue that they use IRR to establish a comparison in analysis outcomes between manual and AI-assisted coding, in our opinion, this practice is still counted as a quality assurance strategy that helps validate the codes/themes produced by GenAI. From a Big Q perspective, the themes only validate or are trustworthy when they are produced by the researchers' positionality, subjective judgement, and reflexive processes. As such, no matter how good themes and codes produced by GenAI, they are still not researchers' Big Q outputs. The mentioned studies should have avoided methodological incongruence if they did not claim they followed the Big Q TA.

The mentioned practice of using GenAI/LLMs in TA constitutes a clear case of what Clarke et al. (2024) identify as incongruence. The researchers claim to uphold the values of rigorous, grounded qualitative inquiry (e.g. reflexive TA, grounded theory), yet their actual practice, forcing a probabilistic language model to act as an autonomous, reflective human coder, fundamentally contradicts those values. Instead of working with LLMs to develop their own deeper interpretation, LLMs are expected to do all the jobs.

2.2.2. Coding is not reduced to countable units

A first problem already occurs when deciding what counts as a 'codable unit', raising questions about how meaning is preserved – or distorted – when data are divided for analysis. This is not an issue when practicing small-q qualitative analysis rooted in positivist or post-positivist values (Nicmanis & Spurrier, 2025); however, when applying methodological frameworks in the Big Q/medium Q, or non-positivist traditions, data segments to be coded are not pre-determined. In non-positivist traditions, segment length varies greatly: a single word, a group of words, a sentence, a half, or a full paragraph might be coded. With verbatim transcripts, sentence boundaries are often unclear. A code is not just a descriptive label (e.g. a noun or a noun phrase; see Saldaña, 2016) but can be interpretative using theoretical groundings and methodological guidance (e.g. axial, focused, and theoretical coding in grounded theory; latent coding in reflexive TA). A code does not just appear as a description of 'what is the topic' but can be diversified to capture diverse nuances such as emotion coding, process (action) coding, value (beliefs, attitudes, values) coding, versus coding, casual coding (Nguyen-Trung, 2025; Saldaña, 2016). As the adopters of positivist approaches applying genAI/LLMs, they often treat coding as a mechanical labelling task, rather than a reflective, interpretive process. In non-positivist traditions, coding is not just an act of labelling or naming; rather, they are an active and subjective act of interpretation, meaning-making, and construction. This means, when working with genAI, human researchers are still the ones who make analytical decisions and interpret the data. They do so by ensuring that the interpretations are grounded in their experience, positionality, theoretical lenses, research methodologies, and the data.

GenAI/LLMs are not specifically designed for qualitative research purposes. When LLMs are integrated into traditional qualitative analysis software such as MAXQDA, ATLAS.ti, or NVivo, AI-assisted coding is typically implemented through a sequential workflow. The model processes the material according to the units it is given and does not independently engage in the recursive movement that characterizes human qualitative analysis. A human analyst may encounter a new idea in segment 10 and then return to segment 3 to revise an earlier coding decision. The LLM does not automatically

perform this kind of back-and-forth comparison unless the software or the researcher explicitly builds such review steps into the workflow.

A chatbot, when asked to 'code' a dataset, generates an output that looks like coding, but it does not necessarily create a coded dataset. Unless researchers' iterative guidance is in place (see for instance, Nguyen-Trung, 2025), unless the data are provided in a structured format, such as an Excel table with one response per row, and unless the output preserves that structure, the chatbot does not work segment by segment at all. It may instead compress, summarize, sample, or privilege the most accessible parts of the material before generating codes, themes, or a report.

De Paoli (2024) tried to better simulate the human coding process by introducing an iterative reduction step. After coding the first interview, the LLM is instructed to compare the codes generated for the second interview against the existing cumulative codebook and to flag codes that 'convey a very similar idea or meaning' so they can be merged or discarded. This procedure removes duplicates, but it still relies on surface similarity between code labels and short descriptions. A human researcher would not collapse codes on that basis alone. They would return to the underlying data, check whether two codes genuinely refer to the same analytic phenomenon, and make decisions grounded in interpretation rather than string-level similarity. In De Paoli's workflow, the consolidation process is driven by the model's comparison of code names, not by an examination of meaning in context, and therefore departs from how analytic judgment is exercised in qualitative research. In working with LLMs, researchers need to be responsible for this iterative revision, recoding, reorganisation, refinement, and reinterpretation. This avoids treating analysis as simple data reduction, where material is cut down to a manageable scale, and instead supports data condensation, where data are compressed while retaining their nuance and analytic meaning (Miles et al., 2014).

To address these issues, some researchers have developed human-in-the-loop systems, such as Anakok et al. (2025). Applying Braun and Clarke's (2006) approach, their process begins with the LLM generating summary points for each paragraph of data, which the researcher must verify. These summaries are then clustered using traditional natural language processing (NLP) techniques, and the LLM assigns initial codes to the clusters. At each stage, the human researcher checks and refines the output. Once coding is complete, clustering is repeated to build themes. Cosine similarity calculations are then used to match summary extracts to these themes. The researcher's role at this stage is to correct mismatches. The final output is a list of grouped themes under higher-order headings, along with the model's proposed conceptual framing. This list then serves as the starting point for finalizing the codebook.

This workflow is technically complex and multi-stage, yet it diverges sharply from Braun and Clarke's vision of TA. Instead of deep interpretive engagement, theme development is reduced to mechanical grouping and labelling. Already the first step, creating summary points is problematic for qualitative researchers working within a non-positivist tradition because it risks losing nuance. In manual coding, non-positivist researchers do not just read the summary points, they need to read the whole dataset to 'experience' the interviews as if they were in the real interviews. By this way, they

master their understanding of the interviews, which will affect how they interpret the data in later stages. In non-positivist data analysis, the researchers' cumulative, reflexive and iterative workflow are central to revising their theming and interpretation. These features are not something we can outsource to GenAI/LLM. When small q researchers attempt to do so via establishing automated or AI agentic workflows, they often fail to align their analysis process and outcomes with the core values of the non-positivist analytical approaches they aim to support.

2.2.3. The theming illusion: topics are not themes

Reflexive TA defines a theme as a *pattern of shared meaning united by a central organising concept* across the dataset (Braun & Clarke, 2022). LLMs, lacking positionality and lived experience, generally produce domain summaries or topic buckets (Morgan, 2023; Nguyen-Trung, 2025). For example, an LLM might group data under 'Managerial Challenges,' identifying *what* was said but fail to interpret the *meaning* attached to those challenges. Even if LLMs can generate what is generally referred to as themes, these are not yet themes in the sense of reflexive TA. Themes generated by LLMs are often descriptive, appearing as a noun or noun phrases (e.g. culture, community, family history, natural beauty) (Nguyen-Trung, 2025). LLM-generated themes therefore remain hypothetical at best: they are suggestions, assumptions, or possible explanations. They still require human interpretation before they can become themes, understood as constructed patterns of latent meaning shaped through the researcher's theoretical lens, methodological judgement, and experience. By presenting these mathematically derived topic summaries as 'themes', researchers are not only misapplying the method but actively stripping the analysis of its interpretive core.

3. Rationales for moving toward a new methodological framework

Our review of both general-purpose chatbot experiments and more sophisticated AI pipelines suggests that GenAI is poorly suited to traditional qualitative coding, at least from the perspectives of Big Q/non-positivist researchers. LLM-generated outputs (codes, clusters, categories, themes) tend to reflect semantic patterns, grouping text based on surface-level similarity rather than deeper meaning. In contrast, human data analysis is deliberately slow and iterative, allowing insight to emerge through sustained engagement with the data. Forcing an LLM to mimic this human workflow without human interpretation introduces unnecessary friction: the limited speed gains are offset by the need for verification and correction; the process risks losing nuance and interpretive depth; outputs may look structured but lack grounded meaning-making.

In short, replicating the appearance of qualitative analysis through automating the coding process does not reproduce its essence. To understand why these replications fail, we must ask a fundamental question: Why do we code data in the first place?

The answer lies in human cognitive limitations. Over the past 100 years, coding has emerged as a way to organize, manage and analyse qualitative data because we cannot hold 20 h of interview data in our working memory simultaneously (Palmer, 1928). We must break data down into manageable units, label them (Seidel, 1998), and physically or digitally retrieve them to find patterns. Coding is a human crutch – a scaffold built to support our limited processing power.

GenAI does not share these limitations. An LLM can process multiple interviews holistically. It does not need to tag a sentence with 'Family Conflict' as an intermediate storage device. Within its context window, it can relate semantically similar passages without first converting them into a human-readable code label. By forcing LLMs to execute line-by-line, or segment by segment, coding of the whole dataset, we are engaging in a fallacy: imposing a workflow designed to accommodate human biological constraints onto an LLM system. An LLM does not need to engage in the laborious process of highlighting and labelling individual data segments for later retrieval and synthesis. Nor does it need to iteratively sort and group codes to construct themes. LLMs bypass the linear, cumulative approach that human researchers must follow. If coding is a workaround for human limits, then the 'solution' to using GenAI in qualitative research is not to build better coding bots, but to remove the crutch. We propose a shift away from 'AI as Coder' toward 'AI as Interlocutor' (Hayes, 2025).

In contrast to the above mentioned trend of mechanical reproduction, a divergence is emerging. Beyond the dominant coding paradigm, a second body of work seeks to align the unique capabilities of LLMs with qualitative values. Morgan (2023) noted early on that 'querying with' GenAI threatens the dominance of manual coding. He then proposed Query-Based Analysis (QBA) (Morgan, 2025) based on iterative TA and abductive reasoning to run reflexive dialogues with AI to construct themes. Friese (2026) proposes Conversational Analysis to the Power of AI (CA^{AI}) – a framework that realigns analytic practices with the actual capabilities of the technology. This framework is methodologically agnostic – it does not prescribe a single 'method,' but rather establishes the boundary conditions for how to analyse qualitative data with LLMs without falling into the trap of replicating coding. From this, she points to a broader paradigmatic shift if GenAI is carefully incorporated in QDA (Friese, 2025).

Drawing on Big Q principles and medium Q practical guidance, namely framework analysis and matrix analysis – two common thematic analysis approaches, Nguyen-Trung and Nguyen (2026) proposed Narrative-Integrated Thematic Analysis (NITA) to deploy LLMs as an assistant in supporting theme construction throughout iterative six stages. This framework allocates the first two stages to planning, designing, preparation for a responsible and ethical use. These frameworks position the LLM not as a coding automaton, but as a conversational partner. This aligns with broader arguments by scholars such as Hayes (2025) and Phan and Le (2025), who similarly advocate for a shift away from mechanical extraction toward 'conversing' with data.

In short, our review of the field's two types of incongruences highlights a critical gap: on one hand, qualitative researchers have not acquired sufficient understanding of the new technologies (technological incongruence); on the other hand, technological adoption lacks in-depth understanding of qualitative research foundations (qualitative methodological incongruence). In reality, these two types of incongruence are two intertwined tensions, which are not in conflict but mutually constitutive. Researchers need to balance and draw on the strengths of both poles (technological literacy and qualitative research literacy) rather than choosing only one. What is needed for the field now is deeper reflection on and the development of methodologically congruent frameworks that could overcome the two incongruence types.

4. Key tenets of methodologically congruent framework

To resolve the current state of methodological incongruence, we argue that researchers must move beyond attempting to automate twentieth-century qualitative coding procedures with twenty-first-century tools. Instead, we must rethink our methods within our new methodological frameworks – and arguably our epistemology – to understand how we create knowledge *with* LLMs, rather than just through them. In the section that follows, we propose some broad tenets that would ensure that incorporating GenAI in QDA is still 'meaningful coherence' (Tracy, 2010) and 'fidelity to the spirit of qualitative work' (Sandelowski, 1993, p. 2). These tenets are not doctors' prescription; they are value-based principles, which are clear enough to mark out what counts as methodological congruence with GenAI and what does not, yet broad enough to leave room for creativity and adaptation. In our opinion, for qualitative research, ensuring adaptation and innovation is key to its success.

4.1. Building on a relational epistemology

In developing reflexive TA, Braun and Clarke (2019, 2022) articulated a distinct Big Q approach to thematic analysis, emphasizing that meaning is actively constructed through the researcher's subjectivity. In this view, the researcher's history, values, and theoretical lens are the sole 'engine of insight,' and any attempt to standardize or automate this process is rejected as a slide into neopositivism ('coding reliability') (Jowsey, Braun et al., 2025). For strict adherents of reflexive TA, knowledge generation is an exclusively human, situated act; GenAI, lacking lived experience and reflexivity, cannot 'know' and therefore cannot analyze.

However, working with LLMs compels us to expand this epistemological boundary. We propose a shift from viewing meaning-making as a *solitary* cognitive act to viewing it as a dialogic assemblage (Krähnke et al. 2025; Schäffer & Lieder, 2023). In our framework, we do not cede the authority of interpretation to the machine. Instead, we treat the LLM as a non-human dialog partner, capable of holistic retrieval and pattern recognition. Researcher, in line with the commitments of non-positivist qualitative research, while the generation of material for interpretation becomes a collaborative human–AI process. The AI can help surface the 'what' of the analysis: summaries, quotations, and contrasting perspectives. The researcher remains responsible for the 'how', 'why', and 'so what': contextualizing meaning, developing the explanatory account, and linking the findings back to the original data and their research purposes (e.g., representing participants' voices). This moves us beyond the binary of 'Human = Interpretivist/Machine = Positivist' (Friese et al., 2026; Greenhalgh, 2026). It acknowledges that knowledge can be now produced in the fluid exchanges and iterations between the probabilistic retrieval of the LLM and the reflexive construction of the researcher. We are not 'finding' the truth with GenAI; we are using GenAI to provoke deeper, more dialogic, and more multi-faceted human construction of meaning.

This epistemological shift raises a defining question: Can this relational approach still be called reflexive qualitative research? If we adhere strictly to reflexive TA as branded by Braun and Clarke (2019, 2022), the answer is likely no (Jowsey, Braun et al., 2025). Braun and Clarke prioritize the researcher's unmediated immersion in the data, viewing the

labor of manual coding as essential to the development of deep familiarity. From this orthodox perspective, introducing an AI intermediary disrupts the solitary connection between analyst and text, rendering the process methodologically incongruent with their specific definitions.

However, as suggested by Nguyen-Trung and Nguyen (2026), we propose distinguishing between Braun and Clarke's specific formulation of reflexive TA and the broader practice of conducting thematic analysis reflexively. The latter refers to any non-positivist thematic approach, including Big Q and medium Q according to Braun and Clarke's (2022) classification, that rejects positivist 'truth finding' in favor of constructed meaning, regardless of the specific tools employed. Under this broader definition, the use of AI is methodologically congruent, provided the epistemological shift to 'relational subjectivity' is explicitly declared. As Friese et al. (2026) argues, reflexivity is not a state of cognitive isolation. We already 'think with' literature, diagrams, and peers. Extending this to 'thinking with' an LLM does not violate the core value of reflexivity, which is to critically interrogate how knowledge is made. It simply expands the network of actors involved in that making.

Therefore, we argue that CA^{AI}, NITA, and QBA represent an evolution of the reflexive tradition. They share the values of reflexive TA (subjectivity as resource, meaning as constructed) but reject the procedural dogma that analysis must be a solitary, manual act. By shifting our epistemology to view knowledge as a dialogic assemblage, we maintain congruence not by following old rules but by upholding the underlying principles of interpretive rigor in a digital age.

4.2. Ethical, responsible and rigorous use

As Paulus and Lester (2024) note, digital tools are not only adapted to support existing methods, but they may also fundamentally change our methodologies and knowledge construction. The path to congruence lies in admitting that while the *values* of qualitative research (depth, nuance, context) remain eternal, the *practices* (traditional coding, segmentation) are not. Integrating GenAI into qualitative analysis, however, requires more than a procedural update; it demands a distinct epistemological, methodological, and ethical shift. We must reconsider the locus of ethical and responsible meaning-making.

The use of GenAI in qualitative research raises ethical concerns around data ownership and rights, data privacy, transparency, interpretive sufficiency, model bias, and the researcher's responsibility and agency (Davison et al., 2024). Addressing these concerns requires ethical safeguards and mitigation strategies across the entire research cycle so that GenAI use remains methodologically coherent rather than an add-on to the analysis (Nguyen-Trung & Nguyen, 2026).

This begins with the design of an appropriate workflow that defines where, why, and how GenAI will be used in relation to the chosen methodology. It also involves selecting tools or models that provide suitable security and privacy protections, such as purpose-built GenAI-supported CAQDAS or dedicated AI platforms for research. Where required, researchers need to seek ethics approval in line with institutional procedures,

develop safeguarding strategies for GenAI use, inform participants clearly about how AI will be used and what risks may be involved, and obtain informed consent.

During analysis, ethical GenAI use requires researchers to become familiar with the dataset, prepare AI-ready data through anonymization or pseudonymization, adapt the GenAI system to the research purpose, and design prompts that are purposeful, contextual, and methodologically aligned. AI outputs should not be accepted at face value. They need to be critically verified, questioned, and reflected on through iterative engagement with the data, the research objectives, and the analytic process. This may require adapting objectives, prompts, settings, or workflow decisions as the analysis develops.

Finally, researchers need to document how GenAI was used through monitoring, evaluation, verification, and learning procedures, including an audit trail and reflexive journal or diary. Reporting should disclose the use of AI, explain its role in the research process, and articulate the researchers' positionality and reflexivity in relation to both the substantive topic and the technological tools used (Nguyen-Trung & Nguyen, 2026; Wheeler, 2026).

4.3. Researcher leads and GenAI assists

Although these approaches (CA^{AI}, NITA, and QBA) differ in their specific steps, they share three core methodological pivots that ensure the researcher remains the interpretive lead while leveraging the AI's computational power. Positioning GenAI as an 'interpreter' would lead to methodological incongruence as GenAI/LLMs are probabilistic text predictors, not social actors with lived experience. Therefore, in a congruent workflow, the division of labour is strict: The AI assists; the Human interprets.

The AI acts as a 'high-powered lens' or 'arduous assistant'. It retrieves relevant passages, summarizes content, and identifies surface-level patterns (topic summaries). 'Themes' identified by GenAI are not yet themes in qualitative research's sense. For example, Morgan (2025) notes that while GenAI can list 'Financial Considerations,' the researcher must decide if that topic constitutes a meaningful pattern. This leads to the attempts to construct patterns of meaning recurring in the dataset to answer the question of how and why finance is considered by participants. The AI then plays a role of invoking and stimulating critical thinking and deeper reflection.

The researcher's role is to lead, decide, and interpret. To maintain this position, we propose a 'human-designed-and-managed loop' – adapted from the 'AI-in-the-loop' approach (see Wise et al., 2026). In this approach, interaction with GenAI is embedded within the broader analytic process. Researchers do not simply submit prompts, receive outputs, and verify them afterwards. Instead, they design and manage the 'loop', prepare the analytic direction, engage critically with each AI-generated response, question or reject inadequate suggestions, ask follow-up questions, return to the data, and progressively develop their own interpretation.

This requires researchers to enter the interaction with a clear understanding of their research questions, theoretical resources, methodological commitments, data materials, and analytic aims. These inputs shape which and how GenAI is used and what kind of support it can provide. Verification, reflection, and adaptation are not separate post-processing steps, but continuous activities within the interaction itself. Researchers need to assess whether an AI-generated summary, contrast, theme candidate, or interpretation

is sufficiently grounded in the data, whether it fits the research question, whether it overlooks relevant tensions or negative cases, and whether it requires further interrogation. In short, GenAI is brought into the 'loop' and used in ways designed and directed by qualitative researchers.

Ideally, this workflow is supported by purpose-built tools for AI-assisted qualitative analysis. Such tools can integrate access to the source material, interaction with the AI system, traceability of outputs, reflexive note-taking, analytic archiving, and documentation of decisions within one research environment. This matters methodologically because interpretation does not occur only after an AI output has been produced. It occurs *during* the interaction: when the researcher challenges a response, adds contextual or positional information, reformulates a question, follows up on a surprising pattern, or asks the AI system to return to the data in support of an emerging interpretation.

However, not all researchers have access to purpose-built tools because of limited financial resources, insufficient technical expertise, or inadequate hardware infrastructure. When general-purpose chatbots are used instead, researchers need to compensate for their limitations by creating a parallel analytic workspace (see, for instance, the PERFECT procedure proposed by Nguyen-Trung & Nguyen (2026)), such as a research diary, document, spreadsheet, or project log. This space should be used to store relevant outputs, record prompts and follow-up questions, document decisions, note doubts and disagreements, compare alternative responses, and develop the researcher's own interpretation. This external workspace is not merely an administrative add-on; it becomes a thinking tool – necessary because general-purpose chatbot interfaces are not usually designed for qualitative research. These chatbots may provide weaker grounding in the source material, limited traceability, uncertain auditability, and data security conditions that must be checked carefully before use.

In both cases, GenAI provides suggestions, summaries, contrasts, retrieval support, and possible lines of questioning. The researcher remains responsible for assessing these contributions, situating them within the study's theoretical and methodological frame, documenting analytic decisions, and constructing the thematic narrative or interpretation. The AI may assist in surfacing the 'what' of the analysis, but the researcher develops the 'how,' 'why,' and 'so what' through sustained engagement with the data, the research context, and the purpose of the study.

4.4. From mechanical extraction to dialectic and abductive discovery

In traditional coding, researchers fragment data to manage it. In the mentioned frameworks like CA^{AI}, NITA, QBA, researchers leverage the LLM's ability to process data abductively and dialectically.

The key shift here is twofold. First, we shift from line-by-line coding and tagging segments by segments to a dialogic mode of deep thinking and questioning. This mode emphasises leveraging the actual strength of LLMs – not coding automation, but enabling a reflective, interactive conversation about the data. Researchers can begin by prompting the LLM to extract initial themes, guided by contextual information such as research questions or theoretical framing. They can then interrogate those themes, ask follow-up questions, extract supporting quotes, and verify answers directly against the original source data (Morgan, 2025) – as is possible with some GenAI tools specifically designed

for qualitative analysis. Because the researcher remains involved throughout the process, every LLM response is critically reviewed, unproductive prompts can be revised, and the analysis can evolve through ongoing dialogue.

Second, we shift toward an abductive and dialectical process of intellectual engagement with GenAI. This process may not need to begin with a priori theoretical concepts or fixed analytic categories. In an initial thematic analysis, GenAI can be used exploratively to surface candidate themes, recurring topics, contrasts, tensions, or surprising patterns directly from the data. These suggestions are not yet findings or fully developed themes; they are provisional analytic material for further interpretation. The abductive work begins when researchers question these suggestions: why a pattern matters, how it relates to the research question, whether it is sufficiently grounded in the data, what alternative interpretations are possible, and whether negative cases or contextual differences challenge the emerging account. Theoretical concepts may be introduced to guide, focus, or interrogate the analysis, but this is a methodological choice rather than a necessary starting point. The central shift is therefore toward a dialogic process in which candidate meanings are generated, questioned, compared, and progressively developed into a researcher-led interpretation. The analysis continues until researchers have developed a rich interpretation of the data that helps answer their research questions. This abductive, dialogic process preserves the iterative, recursive nature of qualitative work – the 'deep immersion' – without the mechanical burden of manual coding. The analysis evolves through conversation, not accumulation.

5. Conclusion: from replication to methodological evolution

The integration of GenAI into qualitative research represents both opportunities and challenges. One of the challenges comes from the fact that its current application and trajectory are often methodologically incongruent and even flawed. As we have argued, the prevailing trend of using LLMs to automate coding and data interpretation represents a skeuomorphic fallacy: a retreat into familiar habits that paradoxically undermines the interpretive depth we seek to preserve. By forcing a probabilistic, holistic reader (GenAI) to replicate human coding, researchers impose a deterministic, linear constraint on a process that should be iterative and fluid. Thus, researchers create the very incongruence they fear – producing analyses that are mechanistically rigid yet methodologically shallow. In this way, qualitative researchers could dig their own hole and bury their identity as a qualitative investigator.

The solution is not to outright reject the technology nor adopt it uncritically for the sake of speed and efficiency. The solution is to evolve our methodology while maintain and grow our reflexivity and interpretation. By adopting frameworks such as CA^{AI}, NITA, and QBA, researchers can actively develop newly methodological approaches that align their choice of technologies and AI-assisted practices with their values and epistemological lens. This requires a pivot from partitioning data to interrogating patterns and from viewing the researcher as a solitary coder to an interpreter in relational socio-technical ecosystems of knowledge production.

Ultimately, achieving methodological congruence in the age of AI requires qualitative researchers to trust, rather than minimize, their interpretive subjectivity. If GenAI is

understood as a powerful system for retrieving, summarizing, comparing, and suggesting possible patterns, then the burden of meaning-making remains firmly with the human researcher. We must move beyond the anxiety of 'replacement' and embrace the researchers' responsibility as bricoleur.

The qualitative researcher as bricoleur (Klag & Langley, 2013) is a skilled maker who works creatively with the resources available to them: the data, theoretical concepts, methodological commitments, their own assumptions and experience, colleagues' perspectives, and GenAI-generated suggestions. These resources do not determine the interpretation; they provide material for analytic work. The researcher's task is to navigate tensions, contrasting ideas, and assumptions, and then synthesize them into an integrative interpretation that connects the data to the research questions. In doing so, we do not surrender the 'soul' of qualitative research. Rather, we loosen the dependence on labor-intensive coding as the route to interpretation, allowing researchers to focus more directly on questioning, comparing, theorizing, and constructing meaning from the data.

Author contributions

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Summary of paper and contribution

This paper shows why automating qualitative coding with Large Language Models (LLMs) results in methodological incongruence. We advocate for a shift to 'relational' meaning-making, demonstrating how this approach remains congruent with the principles of reflexive qualitative research.

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